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Inventor(s)	Wright; Mark Edwin

Portable liquid cover system

Abstract

A liquid cover system comprising a vessel, a brush, and a hose, which efficiently applies liquid cover to a surface. The system includes an upper chamber and a lower chamber separated by a flexible diaphragm, which are pressurized by a handle connected to a pump body to force liquid cover through an outlet and the hose to the brush. The brush includes a pliable tube held at a slight angle towards one edge of bristles, to efficiently apply liquid cover to trailing edge bristles. The brush is secured to a handle, which includes a lever for controlling paint flow and a threaded outlet, and a sliding dovetail connection provided by a top joint and a bottom joint.

Inventors: **Wright; Mark Edwin (Conesville, OH)**

Applicant: **Wright; Mark Edwin** (Conesville, OH)

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Primary Examiner: Walczak; David J

Background/Summary

BACKGROUND OF THE INVENTION

(1) The present disclosure relates to a novel device for applying paint or other liquid covering material onto a surface via a brush.

SUMMARY OF THE INVENTION

(2) Broadly, the invention is a portable liquid cover system, such as for painting, providing mobility and safety to the person using it. The invention can be used to apply a variety of liquid cover materials to surfaces, such as varnish or stain for finishing wood, ink for large scaled calligraphy or drawing such as artwork for murals, or adhesives. It can also be used in various industrial applications, such as coating metals or plastics with lubricants or other liquids.

(3) The invention includes a hand air pump screwed into the top portion of a vessel. The top part is cinched with the lower paint chamber, which has a rubber bladder to keep the air and paint separated. Pumping the air pump pressurizes the rubber bladder, forcing the paint in the lower paint chamber through a tube at the bottom of the chamber to the hand valve.

(4) The hand valve has a machined and a molded end that can accommodate paintbrushes of different sizes by sliding them together. Once attached, a screw or washer secures the brush in place. When the hand valve lever is depressed, it regulates the amount of paint flowing to the bristles through a pliable, flexible tube, which moves with the brush in any direction. This pliable tube can be made at different lengths along the brush bristles to cater to various applications, be it painting walls or ceilings. The vessel, along with the hose and paintbrush, can be carried in a backpack. The backpack features a zipper area or slot to accommodate the hose connection. A magnet or spring-loaded clip across a front panel of the backpack, in the chest area, secures the

paintbrush in an upright position to prevent dripping. This provides significant safety for the person using this, allowing the user to have both hands free to move about and climb a ladder safely.
(5) The materials used to make this invention include plastic vessel, rubber air pump, plastic hand valve, machined aluminum for sliding together to change the brush, steel tubing, wooden paintbrush, rubber balloon, and screws.
DETAILED DESCRIPTION OF THE INVENTION
(6) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof, and in which are shown various specific embodiments in which the disclosure may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, and it is to be understood that other embodiments may be utilized and that mechanical, procedural, and other changes may be made without departing from the spirit and scope of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense. The scope of the present disclosure is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.
(7) The following reference numbers are used in the figures and accompanying descriptions:
(8) The following reference numbers are used in the figures and accompanying descriptions:
(9) TABLE-US-00001 No. Name 100 liquid cover system 102 brush 104 hose 106 vessel 108A pump handle 108B pump body 110 top chamber 112 bottom chamber 114 latch 114A top latch 114B bottom latch 116 outlet 118 connector 120 diaphragm 122 liquid cover 200 brush handle 202 lever 203 outlet 204 top joint 204A input hole 204B groove 206 screw 208A set screw 208B washer 210 bottom joint 210A tongue 212 rigid tube 214 pliable tube 216 binder 218 bristles 220 “O” ring 302 person 304 backpack 306 front panel 308 back compartment

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Description

BRIEF DESCRIPTION OF DRAWINGS
(1) For a fuller understanding of the nature and advantages of the present invention, reference should be had to the following detailed description taken in connection with the accompanying drawings, in which:
(2) The figures here represent one embodiment of the invention.
(3) FIG. **1** is a view of the completed liquid cover system, comprising a vessel, hose, and brush.
(4) FIG. **2**A shows various important components of the assembled vessel.
(5) FIG. **2**B is an exploded view, principally showing chambers and a diaphragm between them.
(6) FIG. **2**C is an interior view of the bottom chamber.
(7) FIG. **3**A is an exploded view of the brush.
(8) FIGS. **3**B and **3**C are side views, respectively, bottom joint and top joint.
(9) FIG. **3**D shows the assemble brush.
(10) FIG. **4** is a cut away view of the assembled brush.
(11) FIGS. **5**A and **5**B show a person using the liquid cover system with a backpack.
DETAILED DESCRIPTION OF THE DRAWINGS
(12) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof, and in which are shown various specific embodiments in which the disclosure may be practiced. These embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, and it is to be understood that other embodiments may be utilized and that mechanical, procedural, and other changes may be made without departing from the spirit and scope of the present disclosure. The following detailed description, therefore, is not to be taken in a limiting sense. The scope of the present disclosure is defined only by the appended claims, along with the full scope of equivalents to which such claims are entitled.
(13) FIG. **1** shows a liquid cover system **100** as assembled, including a vessel **106**, a brush **102**, and

a hose **104** connected to the vessel **100**.
(14) FIG. **2**A identifies key parts of the liquid cover system, including vessel **106** having an upper chamber **110**, lower chamber **112**, latches **114** that hold the upper **110** and lower **112** chambers together, outlet **116**, hose **104** delivering liquid cover to the brush, and connector **118**. Pump handle **108**A is used in conjunction with pump body **108**B (discussed below) to create air pressure in the upper chamber. The shape of pump handle **108**A and its attachment to pump body **108**B show that the pump handle **108**A is arm operated, by person **302** (discussed below with respect to FIG. **5**A), as would any pump with a handle similar to pump handle **108**A.
(15) FIG. **2**B is an exploded view. Diaphragm **120** is deployed between upper chamber **110** and lower chamber **112**, to separate liquid cover in the lower chamber from air in the upper chamber in an air-tight manner. The pump handle **108**A is pumped to cause pump body **108**B to create air pressure in the upper chamber **110**. The diaphragm **120** is flexible so that air pressure applied to the upper chamber applies corresponding pressure, without leakage, in the lower chamber to force liquid cover through outlet **116** and through the hose **104**, via secure connection provided by connector **118** attached to outlet **116**. Furthermore, the diaphragm **120** may be disposable so as not need to be cleaned, or of a more permanent design which would requiring cleaning.
(16) FIG. **2**C is an interior view of the bottom chamber, principally showing liquid cover **122**.
(17) FIG. **3**A is an exploded view of brush **102**. Brush handle **200** includes a lever **202** to control paint flow, and a threaded outlet **203**. Brush handle **200** is screwed into top joint **204** and hole input **204**A, with O ring **220** between them. Rigid tube **212** and pliable tube **214** connected to it are inserted into bottom joint **210**. Then the combination of bottom joint **210**, rigid tube **212**, and pliable tube **214** are inserted brush binder **216** and secured by screws **206**, so that rigid tube **212** remains straight along the brush axis while pliable tube **214** departs from the rigid tube at an angle toward a leading edge of the bristles **218**, the outlet of the pliable tube being positioned closer to said leading edge than to an opposite edge of the bristles. This configuration delivers liquid cover **122** to the leading edge, allowing trailing bristles to spread the coating evenly.
(18) Refer now to FIGS. **3**B and **3**C for side views, respectively, of bottom joint **210** and top joint **204**, showing the sliding dovetail attribute with groove **204**B in top joint **204** and tongue **210**A in bottom joint **210**. Top joint **204** and bottom joint **210** are slid into place and alignment, each with holes reaching through to deliver liquid cover **122** through brush handle **200** and eventually through pliable tube **214** and into bristles **218**. This connection between top joint **204** and bottom joint **210** is secured by a pair of set screws **208**A and washers **208**B.
(19) FIG. **3**D shows the final, assembled brush **102**.
(20) For clarification, FIG. **4** is a cut away view of assembled brush **102**, further showing placement and connection of the various components.
(21) FIGS. **5**A and **5**B introduce a person **302** wearing a backpack **304**. The backpack **304** includes front panel **306** and back compartment **308**. Back compartment **308** is designed to hold vessel **106**, and to allow hose **104** to be connected to outlet **116**. Front panel is provided with some means, such as a clip or magnet, to hold brush **102** while not being actively used to apply liquid cover **122** to a surface.
(22) In addition to the advantages of this invention listed above, several ergonomic and efficiency advantages and benefits accrue in a liquid cover system **100** that does not require a person **302** to bend over to get paint on a brush **102**. These include reduced strain on the back. By providing an alternative method for getting paint onto the brush **102**, the person **302** can maintain a more neutral and comfortable position, reducing the risk of back pain and injury associated with prolonged bending.
(23) Another benefit is improved posture. With very limited bending over, a person **302** can maintain a more upright and neutral posture, which can reduce stress on the back and other parts of the body. By keeping the body in a more ergonomic position, the person **302** can help prevent issues such as muscle tension, discomfort, and fatigue.

(24) Yet another benefit is increased efficiency. By limiting the need to bend over, a person **302** can work more efficiently and with less fatigue. This can help reduce the risk of mistakes, risk and danger of spillage, and increase productivity.
(25) Still further, another benefit is better access. The liquid cover system **100** keeps the paint or other liquid cover **122** readily available to apply to a surface, and at reduced risk of spillage. Person **302** can more easily access the liquid cover **122** without having to reach, stretch, or strain. This also reduces the risk of spillage. All of this can help to reduce the risk of slips, trips, and falls, as well as improve overall safety in the workspace.
(26) Although the invention has been described with reference to various embodiments, those skilled in the art will understand that various changes may be made and equivalents may be substituted for elements thereof without departing from the scope and essence of the disclosure. In addition, many modifications may be made to adapt a particular situation or material in accordance with the teachings of the disclosure without departing from the essential scope thereof. Therefore, it is intended that the disclosure not be limited to the particular embodiments disclosed, but that the disclosure will include all embodiments falling within the scope of the appended claims. Also, any and all citations referred to herein are expressly incorporated herein by reference.

Claims

1. A system for applying liquid cover to a surface, comprising a vessel (**106**) having an upper chamber (**110**) and a lower chamber (**112**) separated by a flexible diaphragm (**120**), wherein said flexible diaphragm (**120**) is disposable, latches (**114**) holding the upper chamber (**110**) and lower chamber (**112**) together, an outlet (**116**) for liquid cover (**122**), a hose (**104**) connected to the outlet (**116**), a brush (**102**) adapted to receive liquid cover (**122**) from the hose (**104**) via a pliable tube (**214**), the brush (**102**) further having bristles (**218**), the brush (**102**) further having a brush handle (**200**) and a lever (**202**), wherein the lever (**202**) is manually operable, a pump handle (**108**A) wherein the pump handle (**108**A) is connected to a pump body (**108**B), further wherein the upper chamber (**110**) is pressurized by operation of the pump handle (**108**A) and the pump body (**108**B), wherein the pump handle (**108***a*) is arranged to receive an applied force, which causes pressure in the upper chamber (**110**) to apply corresponding pressure to the lower chamber (**112**) to force the liquid cover (**122**) through the outlet (**116**) and hose (**104**) to the brush (**102**).
2. The system of claim 1, wherein the brush (**102**) further comprises a rigid tube (**212**) attached to the pliable tube (**214**), the rigid tube extending along a longitudinal axis of the brush handle and the pliable tube extending from the rigid tube at an angle toward one edge of the bristles (**218**) such that an outlet end of the pliable tube is nearer to said edge than to an opposite edge of the bristles, thereby delivering liquid cover (**122**) to that edge for efficient spreading by trailing bristles.